BCSE429P

Cyber Physical Systems Design Lab

Laboratory 9 - Audio Clip Player and April Tags using Accessor API

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Question:

Do the following exercises and check the output:
1. Audio Clip Player using Accessors API
2. April Tags for Industrial Machinery
As per the following document.

You need to open capecode application instead of vergil.
In Capecode, you will have two demos: One with Clip Audio Player and other with April Tags.
Download some AprilTags from the internet and scan it using your Mobile Phone camera and check the Machine ID;
Prepare a report for both exercises with demo screenshot and output screenshot and upload it using your registration Number.

1. Audio CLip Player

Code:

```
exports.setup = function () {
    this.parameter('clip', {type: 'string',
        value: '$CLASSPATH/ptolemy/actor/lib/jjs/modules/' +
               'audio/demo/Audio/strum.wav'});
    this.output('clipURL', {type: 'string'});
    this.output('start', {type: 'boolean'});
    this.output('status', {type: 'string'});
};

exports.initialize = function () {
    this.send('clipURL', this.getParameter('clip'));
    this.send('start', true);
    this.send('status', 'Strum clip requested through Accessor API');
};
```

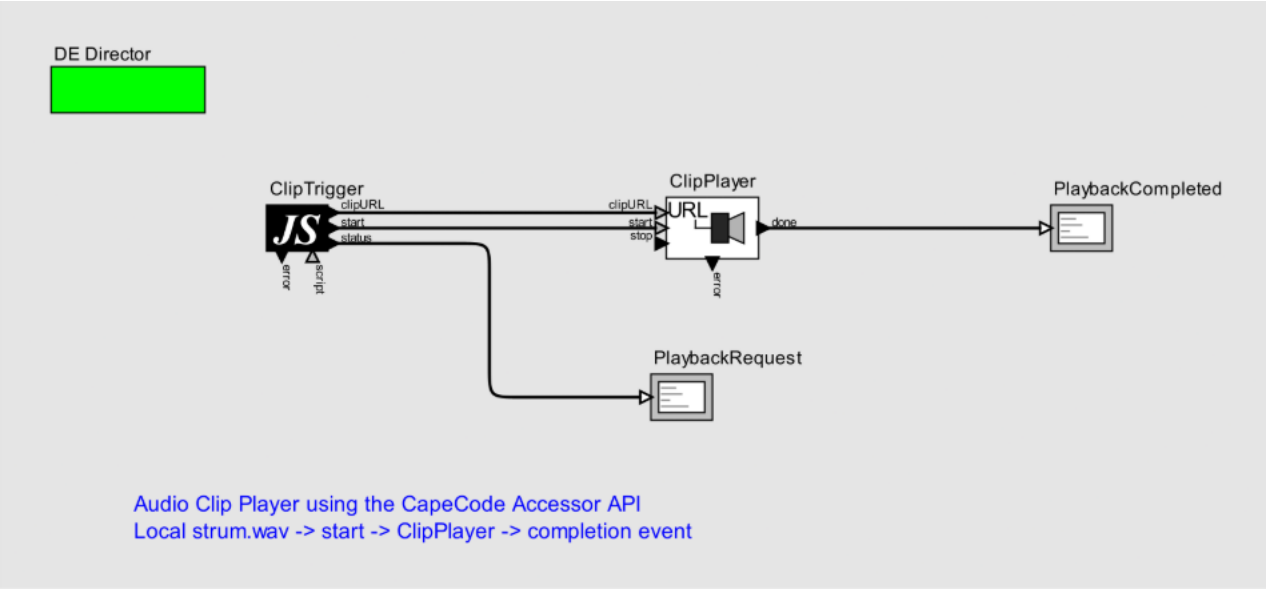
The ClipTrigger sends the local resource URL and a start event to the stock audio/ClipPlayer accessor. ClipPlayer exposes clipURL, start and stop inputs and a boolean done output.

Model settings and connections

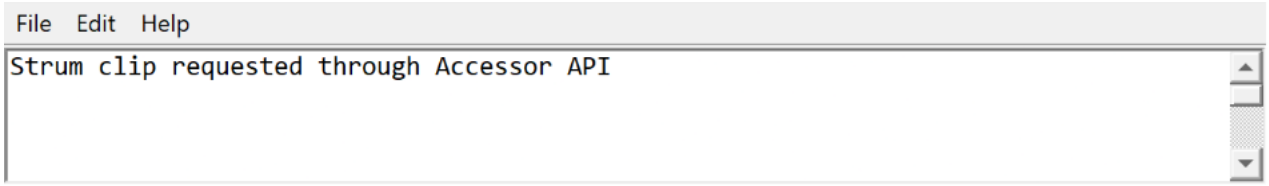
```
DE Director: stopTime = 10 s
synchronizeToRealTime = true

ClipTrigger.clipURL -> ClipPlayer.clipURL
ClipTrigger.start   -> ClipPlayer.start
ClipTrigger.status -> PlaybackRequest.input
ClipPlayer.done    -> PlaybackCompleted.input
```

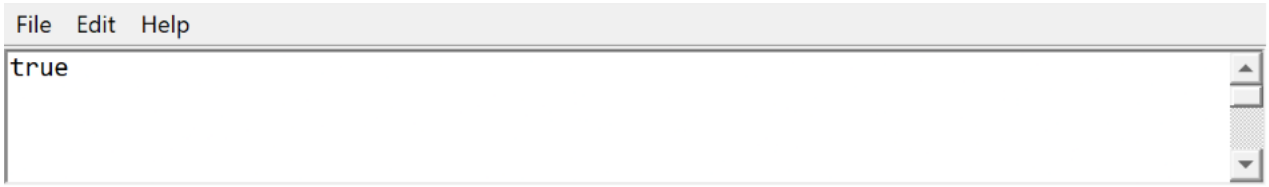
Open AudioClipPlayer.xml in CapeCode and press Run. The complete stock ClipPlayer source is supplied as code/ClipPlayer.js, with its original licence.



Playback request



Playback completion



The actual ClipPlayer emitted done = true. The local strum.wav contains stereo audio at 44,100 Hz and lasts 4.455 s. The model ran to its 10 s stop time and returned to the idle state without an execution error.

2. April Tags

Code:

```
var machines = {19: 'MACHINE_019', 20: 'MACHINE_020', 21: 'MACHINE_021'};

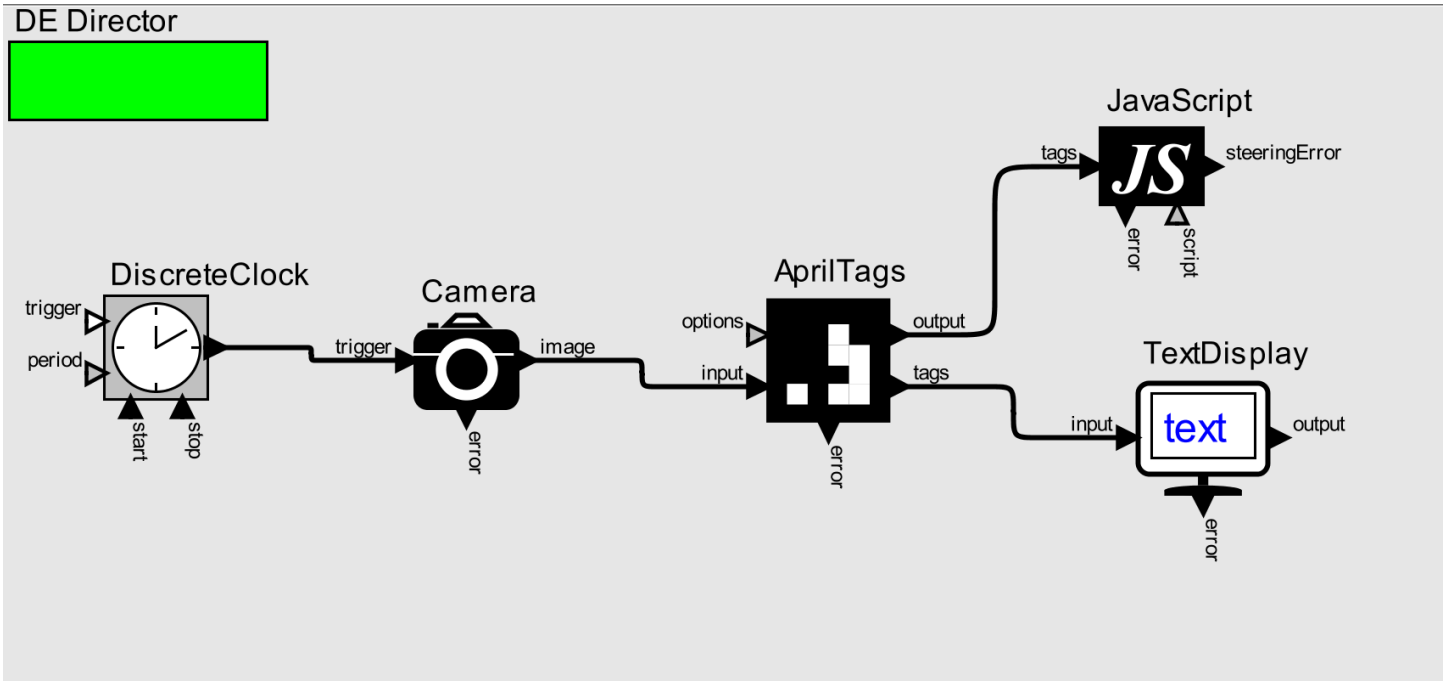
exports.setup = function () {
    this.input('tags');
    this.parameter('frameWidth', {type: 'number', value: 640});
    this.output('machineInfo', {type: 'string'});
    this.output('steeringError', {type: 'number'});
};

exports.initialize = function () {
    this.addInputHandler('tags', function () {
```

```
        var tags = this.get('tags');
        var width = this.getParameter('frameWidth');
        if (!(width > 0)) { throw new Error('Invalid frameWidth'); }
        if (!tags || tags.length === 0) {
            this.send('machineInfo', 'NO_TAG_DETECTED');
            this.send('steeringError', 0);
            return;
        }
        for (var i = 0; i < tags.length; i++) {
            var tag = tags[i];
            var error = tag.center[0] - width / 2;
            var info = {tagId: tag.id,
                        machineId: machines[tag.id] || 'UNREGISTERED',
                        center: [Number(tag.center[0].toFixed(2)),
                                 Number(tag.center[1].toFixed(2))],
                        steeringError: Number(error.toFixed(2))};
            var message = JSON.stringify(info).replace(',"center"', ',\n "center"');
            this.send('machineInfo', message);
            this.send('steeringError', error);
        }
    });
};
```

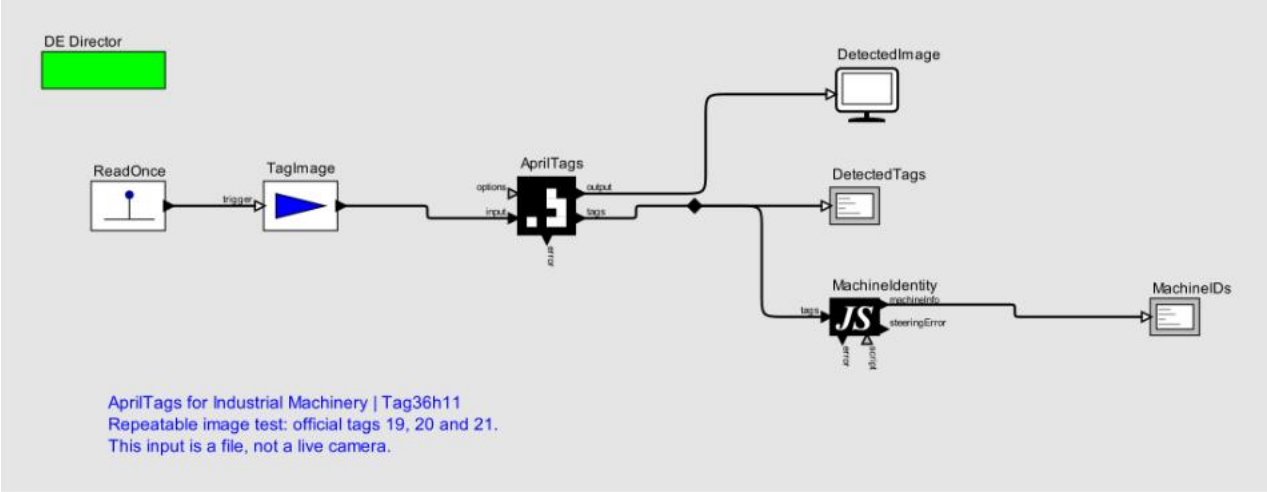
The Camera accessor is configured for 5 frames/s and a 640-pixel-wide view. The repeatable image test uses the same detector and mapping with frameWidth = 1500. Machine names are application-defined mappings; the tag itself encodes a numeric ID.

Steering error = detected centre x minus half the image width. The error is reported only; this exercise does not command industrial hardware.



Camera -> AprilTags -> image display, tag data and machine-ID lookup. This model was opened in CapeCode with maxFrameRate = 5. The physical scan was performed separately on the mobile phone, as shown in the supplied screenshots.

Repeatable CapeCode detection model



The executed test reads the downloaded tag image and uses the same AprilTags accessor. This establishes the detector and machine-ID outputs independently of the phone scanner.

